DESIGN NOTES

LoRa-Based MQTT Bridge (Receiver Side)

Antenna Placement · Power Budget · Production (1,000-Unit) Changes

1. Antenna Placement Rationale

- **Single RF antenna on this board:** the 868 MHz LoRa antenna (E2, ANT-868-PW-RA, tilt/swivel screw-mount whip, feeding RAK3172 U1's RF pin). Being an external screw-mount type rather than a PCB trace or chip antenna gives more placement freedom, but ground-plane extent and feed routing still matter for match and efficiency.
- **Mounting & orientation:** mount E2 at the enclosure edge, vertical, for the omnidirectional pattern typical of gateway/bridge installs. Keep it clear of metal enclosure panels, brackets, or standoffs, and away from the ESP32-DEVKIT-V1's own onboard antenna to avoid pattern distortion and detuning.
- **Coexistence with ESP32 Wi-Fi:** the ESP32-DEVKIT-V1 carries its own onboard 2.4 GHz Wi-Fi antenna a few centimetres from the RAK3172 and its 868 MHz antenna on the same board. The two bands don't overlap, but a strong Wi-Fi TX burst sitting that close to the LoRa RF front end can raise its noise floor and desensitize LoRa RX. Recommend physically separating the two modules on the PCB/enclosure and, if range is marginal, adding shielding or extra spacing between them.
- **UART link routing:** the ESP32–RAK3172 UART bridge (TX2/RX2 to PB6/PB7) should be routed short and kept away from the antenna feed line to avoid coupling RF noise into the digital signal.
- **Enclosure material:** if the gateway is housed in a metal enclosure, the whip antenna must protrude outside it (or the enclosure needs an RF-transparent window) — a fully metal box around a whip antenna will substantially attenuate the LoRa link.

2. Power Budget Estimate

Note: unlike the collar, this schematic shows no battery/charging section — the bridge is assumed to run from a continuous 5 V USB or DC supply feeding the 3.3 V rail shown. The estimate below frames the supply's current headroom rather than a battery runtime.

State	Dominant load	Approx. current
Idle / LoRa RX listen	ESP32 Wi-Fi connected & idle + RAK3172 continuous LoRa RX	~80–120 mA (ESP32 Wi-Fi idle) + ~10–12 mA (RAK3172 RX) ≈ 90–130 mA
Wi-Fi TX burst (MQTT publish)	ESP32 Wi-Fi radio driving an MQTT publish over the WLAN uplink	~150–260 mA (peaks during association / TX)
LoRa TX burst (ack / downlink)	RAK3172 PA drive, if the bridge also transmits, ~14 dBm typical up to +22 dBm boost	~20–40 mA (14 dBm) – ~100–120 mA (22 dBm)
Worst-case concurrent peak	Wi-Fi TX burst overlapping a LoRa TX burst	~300–400 mA combined — size the supply/regulator with margin above this

Takeaway: *unlike the battery-powered collar, this bridge is continuously mains/USB powered, so the real design lever is the current capacity and thermal margin of the (not-yet-shown) 3.3 V regulation stage and supply connector — sized to the ~300–400 mA worst-case concurrent peak above, not battery runtime. If a future revision adds a backup battery for power-outage buffering, the collar's lesson about always-on linear-regulator quiescent current applies equally here.*

3. Changes for a 1,000-Unit Production Run

- **Replace dev boards with bare modules:** swap the ESP32-DEVKIT-V1 breakout for a bare ESP32-WROOM-32(E) module soldered directly to the production PCB. The DEVKIT carries a USB-UART bridge, its own regulator, and headers that duplicate functions on a purpose-built gateway board — removing it cuts per-unit cost substantially at volume.
- **Add a defined power input & regulation stage:** this schematic only shows a bare 3.3 V rail; the production board needs an actual input path (USB-C or DC jack), reverse-polarity/ESD protection, and a 3.3 V regulator (buck preferred over linear for efficiency) sized for the peak Wi-Fi + LoRa current identified above.

- **Retain a programming/debug path:** if the bare WROOM module replaces the DEVKIT, add a small CH340E/CP2102-class USB-UART bridge, or program via a pogo-pin/UART test jig at assembly, mirroring the same trade-off made on the collar's production BOM.
- **BOM cost at scale:** RAK3172 module pricing drops materially at 500–1,000+ unit reel quantities versus the single-unit price used in the current BOM — negotiate reel/tape pricing directly with the module vendor or an authorized distributor.
- **Antenna trade-off:** consider whether the ANT-868-PW-RA screw-mount whip is still the right choice at volume, versus a lower-cost SMD/chip antenna or a pre-terminated pigtail antenna, depending on the enclosure design and the range/gain the install requires.
- **RF certification:** as a device transmitting/receiving on both 868 MHz LoRa and 2.4 GHz Wi-Fi, this bridge needs ETSI (or applicable regional) radio certification for both radios before a production run — schedule this early, since it can drive antenna and layout changes.
- **Enclosure & mounting:** move to an injection-molded, IP-rated enclosure suited to the install location (indoor wall-mount vs outdoor gateway), with an RF-transparent antenna window or an external antenna pass-through per the placement notes above.
- **Status indication:** add status LEDs (power, Wi-Fi-connected, LoRa activity) to aid installation and field diagnosis without needing a laptop/serial console.
- **OTA firmware updates:** for a fixed-install gateway, add Wi-Fi-based OTA update capability so deployed units don't need physical access for firmware updates at volume.